

Frequency-Guided Semantic Steganography for High-Fidelity Self-Contained Style-Transfer Image Hiding

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Abstract—Style transfer steganography conceals data within artistic brushstrokes, but methods in this family typically report cover PSNR in the 33–36 dB range and fail under repeated re-stylisation. These limitations often stem from unbounded capacity penalties that leak embedding energy into low-frequency structures, alongside spatial payload representations that strain the carrier’s capacity. To overcome these issues, we propose a frequency-guided semantic steganography network (FGSS). We first compress the secret into a compact $64 \times 32 \times 32$ semantic token and a thumbnail, reducing bandwidth requirements by roughly a factor of three. We then introduce a nonlinear capacity ceiling that maps a target PSNR to a soft barrier on the cover MSE. To further hide the residual, a frequency loss inspired by the discrete wavelet transform confines the perturbations to the high-frequency band. Finally, we employ an oracle-guided reverse decoder and a probabilistic future pass so that the self-contained stylisation survives up to two further re-stylisations, i.e. three cumulative stylisation passes. Evaluated on 600 pairs from COCO val2017 across five painting styles, FGSS achieves 39.61 ± 0.35 dB cover PSNR and 0.9914 SSIM. It improves reverse SSIM from 0.52 (naive baseline) to 0.81, and maintains token stability under JPEG and Gaussian noise. Ablation studies confirm the capacity ceiling alone provides a 5.76 dB gain. Cross-dataset tests demonstrate consistent ≈ 40 dB PSNR, and robustness checks across 11 attacks show negligible token drift. The embedding path needs only 1.9 M of the system’s 10.7 M parameters and embeds a 256×256 image in under 0.01 s on a consumer GPU.

Index Terms—Image steganography, Neural style transfer, Frequency-domain learning, Capacity ceiling, Robustness, Semantic tokens.

I. INTRODUCTION

Hiding information inside ordinary-looking images is a fundamental technique in digital communication. Deep learning has substantially extended the achievable payload capacity relative to LSB-based predecessors; however, the residuals introduced by deep encoders exhibit statistical fingerprints that contemporary steganalysers detect with considerable accuracy [1], [2]. One countermeasure is to alter the statistical properties of the carrier itself: when the cover is a painting

rather than a photograph, brush strokes and palette quantisation already introduce noise within which small additive perturbations remain visually plausible. This motivates *style-transfer steganography* [3], [4], [5], and in particular its most demanding variant, *self-contained* stylisation [3], in which the stego image alone must carry sufficient information for the receiver to reconstruct the original content by inverting the stylisation process. In practice, balancing these objectives is difficult. Increasing embedding strength improves secret recoverability but introduces visible texture jitter and palette drift; reducing it preserves the cover but degrades reverse extraction; and a second or third style transfer applied to the stego image — serial stylisation — typically destroys most of the embedded information [3]. Consequently, most reported systems occupy a narrow region of this trade-off space, with cover-referenced PSNR typically in the 33–36 dB range, reverse SSIM below 0.6, and serial recovery degrading rapidly beyond the second stylisation step.

This trade-off is driven by two common design choices. First, models typically use an unbounded payload capacity loss: the encoder is encouraged to keep the residual small only through a quadratic prior, which imposes no upper bound, so nothing prevents the optimiser from placing a successful solution in the low-frequency, structurally salient component of the carrier, to which the human visual system is most sensitive. Second, they rely on a spatial payload representation: secret images, dense bit grids and feature maps occupy approximately as many spatial positions as the cover itself, forcing the encoder to distribute its perturbation broadly and leaving little capacity for the carrier to absorb a subsequent stylisation pass. This work addresses both limitations. Instead of a spatial payload, the secret is distilled into a compact semantic token $\mathbf{z} \in \mathbb{R}^{64 \times 32 \times 32}$ together with a low-resolution thumbnail $\mathbf{T}$, both produced by a small CNN tokenizer. Rather than an unbounded prior, the residual is bounded by a nonlinear capacity ceiling: a quadratic penalty that activates only above an explicit threshold ϵ tied directly to a target cover PSNR. A frequency loss based on the discrete wavelet transform further constrains the spectral allocation of the residual, penalising the low band

aggressively and leaving the high band unconstrained, thereby directing the perturbation into texture rather than into structure. Three further mechanisms support the system: an oracle-guided reverse loss, a serial consistency term for robustness to further re-stylisation, and a bandwise embedding allocation loss.

Despite these additions the deployment cost stays low: of the 10.7M trainable parameters in the full system, only the 1.9M in the tokenizer and stego encoder are needed on the sending side, and that path embeds a 256×256 image in 0.0096s on a consumer GPU. On the COCO val2017 [6] protocol with 600 content–style pairs and five painting references, the model achieves a cover PSNR of 39.61 ± 0.35 dB, an SSIM [7] of 0.9914 ± 0.0030 and an LPIPS [8] of 0.0195 ± 0.0081 , while the token MSE varies by at most 1×10^{-4} under JPEG-style smoothing and additive Gaussian perturbations. A controlled CPU-only ablation attributes the improvement primarily to the capacity ceiling, which alone accounts for +5.76 dB of cover PSNR ($35.61 \rightarrow 41.37$ dB). Our contributions are: (i) a capacity ceiling formulation that maps a target cover PSNR to a soft nonlinear barrier on the cover MSE, removing the implicit fidelity/extraction trade-off; (ii) a differentiable frequency-guided loss that uses a strided average-pooling proxy of the discrete wavelet transform to confine the residual to the high-frequency band without adding learnable parameters, plus a band-localised embedding allocation loss; (iii) a semantic token payload that compresses the secret into a low-bandwidth representation so the carrier can absorb it under both constraints; (iv) an oracle-guided reverse extraction stack and a probabilistic future pass that stabilise self-contained stylisation under further re-stylisations; and (v) a controlled study on COCO with 600 evaluation pairs covering reverse and serial recovery under three attack channels, an ablation isolating the ceiling and the frequency loss, cross-dataset generalisation on four unseen benchmarks, an 11-attack robustness audit, GPU and CPU inference budgets, and a steganalysis evaluation with three detectors that quantifies — rather than assumes away — the detectability of the resulting stego images.

II. RELATED WORK

A. Deep Image-in-Image Steganography

The first wave of deep steganography embedded full-resolution images using encoder–decoder or generative models, introducing mechanisms like invertible networks [1] and robust flows [2] to balance PSNR. While effective on photorealistic carriers, small perturbations often shift statistics detectably for steganalysers like SRNet [9]. A second family avoids embedding entirely via semantic tokens (e.g., IDEAS [10]), offering immunity to steganalysis but severely limiting payload bandwidth. The third family, style transfer steganography, uses artistic styles to conceal embedding noise. From early self-contained stylisation [3] to recent frequency-domain [11] and diffusion [5] methods, an explicit barrier on cover distortion remains rare. Most models rely on unbounded quadratic priors. We address this gap using a capacity ceiling derived directly from a target PSNR, while combining

the semantic token intuition of coverless methods with the deterministic carrier generation of style transfer.

III. METHOD

A. Notation and System Overview

Scalars are plain math italic, vectors bold lower case, tensors bold upper case. The cover image is $\mathbf{X}_{\text{cover}} \in \mathbb{R}^{3 \times H \times W}$, the secret content is $\mathbf{C} \in \mathbb{R}^{3 \times H \times W}$ and the painting reference is $\mathbf{S}$. The cover is obtained by an optimisation-based stylisation operator $f_{\text{style}}(\mathbf{C}, \mathbf{S})$ held fixed during stego training. We write $\mathbf{z} \in \mathbb{R}^{d_{\text{tok}} \times g_{\text{tok}} \times g_{\text{tok}}}$ for the semantic token used as payload and $\mathbf{T} \in \mathbb{R}^{3 \times g_{\text{tok}} \times g_{\text{tok}}}$ for the auxiliary thumbnail, with $g_{\text{tok}} = 32$ and $d_{\text{tok}} = 64$ throughout; because the tokenizer ends in an adaptive pooling stage, the same token geometry is produced at both the 256×256 benchmark resolution and the 192×192 CPU resolution.

The system consists of four learnable modules — a semantic tokenizer f_{tok} , a stego encoder f_{enc} , a stego decoder f_{dec} and a reverse decoder f_{rev} — together with a non-learnable attack channel $\mathcal{A}$. All four are trained jointly; Figure 1 summarises the data flow. Training and inference share the same flow. The channel is enumerated over its three modes in both regimes: during training so the decoder sees clean and degraded inputs in a balanced fashion, and at evaluation so every pair is scored under each mode separately.

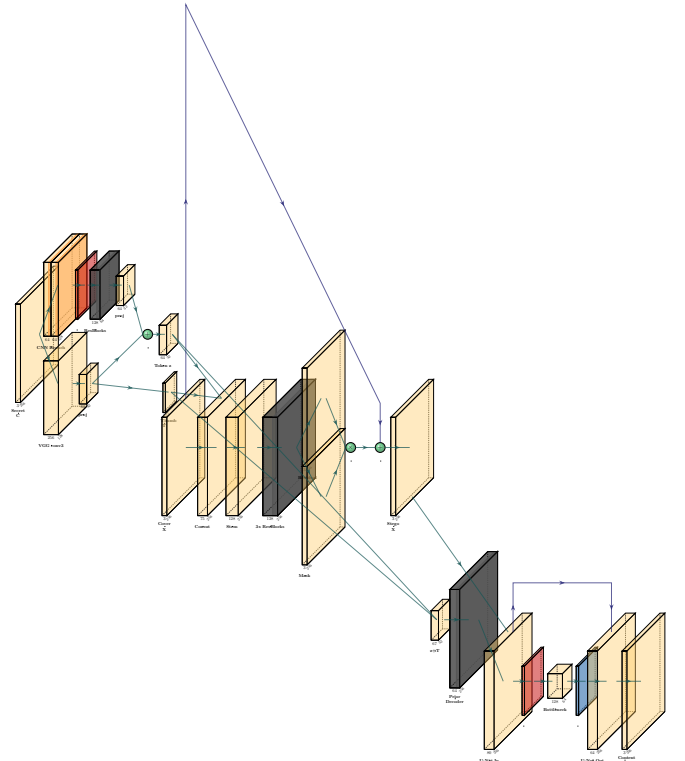


Fig. 1. Frequency-guided semantic steganography network. Green nodes are tensors, blue nodes are learnable modules, orange nodes are differentiable operations, and purple nodes are stochastic but non-learnable. The dashed red boxes mark the three loss families that regulate the joint optimisation.

B. Cover Generation

The cover is computed deterministically from $(\mathbf{C}, \mathbf{S})$ by an optimisation-based operator f_{style} in the spirit of Gatys [12]: a target image initialised at $\mathbf{C}$ is optimised for T_{nst} steps with Adam [15], learning rate 0.03, content weight 1.0 on the `conv_3` feature map of an ImageNet-pretrained VGG-19 [13], and style weight 6×10^5 on the Gram matrices of `conv_1-conv_5`. We use $T_{\text{nst}} = 45$ for the 256×256 benchmark and $T_{\text{nst}} = 120$ for the 192×192 CPU smoke run. f_{style} is frozen throughout stego training and its outputs are cached per $(\mathbf{C}, \mathbf{S})$ pair to keep the inner loop deterministic.

C. Semantic Tokenisation of the Secret

A common pitfall of dense steganography is that the payload is itself spatial, so the encoder must disturb a roughly equal number of pixels. We instead distil the secret. The tokenizer f_{tok} has two parallel branches fused into one token. The *CNN branch* is a strided convolutional network: three $\text{Conv}_{3 \times 3} + \text{GroupNorm} + \text{GELU}$ [14] blocks with stride two (widths $h/2 \rightarrow h \rightarrow h$ with $h = 128$), two residual blocks, and a 1×1 projection to d_{tok} channels, adaptively pooled to $g_{\text{tok}} \times g_{\text{tok}}$. The *VGG branch* extracts the `conv_3` feature map from the frozen VGG-19 backbone [13] and projects it through two 1×1 convolutions ($256 \rightarrow h \rightarrow d_{\text{tok}}$), also pooled to $g_{\text{tok}} \times g_{\text{tok}}$. The outputs are summed and activated with $\tanh$:

$$\mathbf{z} = \tanh(\mathbf{z}_{\text{cnn}} + \mathbf{z}_{\text{vgg}}). \quad (1)$$

In parallel, a bilinearly resized RGB thumbnail $\mathbf{T}$ of the same spatial size is refined through a lightweight $\text{ConvBlock} + \text{ResBlock} + 1 \times 1$ head with sigmoid activation, serving both as a soft anchor for the reverse decoder and as a regulariser that discourages pathological tokenisations. The semantic payload is therefore the pair $(\mathbf{z}, \mathbf{T})$: 68 608 floating-point values ($64 \times 32^2 + 3 \times 32^2$), about three times fewer than the 196 608 values of a $3 \times 256 \times 256$ secret image. We stress what this payload is and is not. It is a *lossy latent* description of the secret, serialising to 33.5 bps in FP32 or 8.375 bps as `uint8` against a 256×256 carrier; it is not a raw message bitstream, and the figures above are therefore a serialisation size rather than a guaranteed lossless capacity. Recovery is also not blind: the receiver needs the trained decoder weights. FGSS consequently targets self-contained stylisation — where sender and receiver share a model by construction — rather than keyless bitstream steganography.

D. Stego Encoder and Decoder

The encoder fuses the cover with four auxiliary feature maps to produce a texture-aware residual. Its input concatenates (i) the cover (3 ch.), (ii) the up-sampled token $\text{up}(\mathbf{z})$ (d_{tok} ch.), (iii) the up-sampled thumbnail (3 ch.), (iv) a two-channel Sobel edge map $\mathbf{E} = \text{sobel}(\mathbf{X}_{\text{cover}})$, and (v) the cover–thumbnail difference $\mathbf{D} = \mathbf{X}_{\text{cover}} - \text{up}(\mathbf{T})$ (3 ch.), yielding 75 input channels. The body is a $\text{ConvBlock}(75 \rightarrow h)$ stem followed by three $\text{ResidualBlocks}(h)$ with two parallel heads: a *residual head* $\mathbf{R} = \tanh(\text{Conv}_{3 \times 3}(\cdot))$ and a *mask head* $\mathbf{M}_{\text{learned}} = \sigma(\text{Conv}_{3 \times 3}(\cdot))$. The learned mask is modulated by a detached

texture gate $\mathbf{G}$, formed by adding the mean absolute Sobel response of the cover to half the mean absolute magnitude of the three high-frequency Haar sub-bands, then normalising by the per-image maximum and clamping to $[0, 1]$; the gate is applied with floor $\tau_{\text{floor}} = 0.6$:

$$\mathbf{M} = \mathbf{M}_{\text{learned}} \odot [\tau_{\text{floor}} + (1 - \tau_{\text{floor}}) \mathbf{G}], \quad (2)$$

$$\mathbf{X}_{\text{stego}} = \text{clip}_{[0,1]}(\mathbf{X}_{\text{cover}} + \alpha \sigma \mathbf{M} \odot \mathbf{R}), \quad (3)$$

where $\alpha = 0.025$ is a fixed residual scale and $\sigma \in [1.02, 1.30]$ at training ($\sigma_{\text{eval}} = 1.24$). Masking steers perturbation energy towards textured regions and away from flat, perceptually salient areas, directly complementing the frequency loss. The trunk width is $h = 128$ at both evaluation resolutions.

The decoder f_{dec} is the dual of the tokenizer-plus-encoder: it consumes a $3 \times H \times W$ stego image, applies three strided ConvBlocks and one ResStack , adaptively pools to $g_{\text{tok}} \times g_{\text{tok}}$, and projects with 1×1 convolutions to $\hat{\mathbf{z}}$ and to a sigmoidal thumbnail $\hat{\mathbf{T}}$ used as a reconstruction anchor. We use a single decoder rather than one per attack mode, for simplicity and to encourage representation transfer across distortion types.

E. Attack Channel and Strength Curriculum

The decoder is supervised under three modes of an attack channel $\mathcal{A}$:

$$\mathcal{A}_{\text{id}}(\mathbf{X}) = \mathbf{X},$$

$$\mathcal{A}_{\text{jpeg}}(\mathbf{X}) = q\mathbf{X} + (1 - q) \text{AvgPool}_{3,1}(\mathbf{X}), \quad q = 0.65,$$

$$\mathcal{A}_{\text{rnb}}(\mathbf{X}) = \text{clip}_{[0,1]}(\mathbf{X} + \sigma_n \boldsymbol{\xi}), \quad \boldsymbol{\xi} \sim \mathcal{N}(0, I), \quad \sigma_n = 0.02.$$

The JPEG-style proxy mimics the low-pass tendency of moderately compressed JPEGs without breaking gradient flow; the Gaussian mode adds zero-mean noise. The strength scalar σ_{train} is sampled per minibatch from $[1.02, 1.30]$ to expose the decoder to a range of embedding magnitudes. All reported test-time results use the fixed value $\sigma_{\text{eval}} = 1.24$, near the upper end of that range. During training we additionally monitor a held-out partition at a slightly stronger $\sigma = 1.36$ for checkpoint selection, so the selected model is chosen under a marginally harder condition than the one it is finally reported at.

F. Loss Functions and Training

To guide the embedding into high-frequency regions where the human visual system is less sensitive, we apply a frequency loss that penalises perturbations in the low-frequency band (extracted via a 2×2 average-pooling proxy) while rewarding energy placed in the high-frequency band. We couple this with a soft capacity ceiling: a one-sided quadratic penalty that activates only when the cover MSE exceeds a threshold $\epsilon \approx 1.2 \times 10^{-4}$ (corresponding to ~ 39 dB PSNR). This guarantees high cover fidelity without forcing the residual to zero.

The reverse decoder f_{rev} reconstructs the original content $\mathbf{C}$ from the decoded token $\hat{\mathbf{z}}$ and thumbnail $\hat{\mathbf{T}}$ using two cascaded sub-networks: a Token Prior Decoder and a U-Net refinement decoder. The reverse loss combines attack-conditioned terms across Identity, Gaussian noise, and JPEG compression modes.

We additionally maintain an oracle distillation signal from a detached ground-truth token to stabilise decoding, and a serial consistency loss that enforces extraction from covers subjected to up to two further re-stylisation passes.

The joint objective combines the capacity ceiling, frequency guidance, reverse extraction, oracle distillation, and serial consistency. Training proceeds in three phases over 41 epochs on an A100 GPU: (1) Fidelity-lock resets the cover model around the target PSNR with damped reverse weights; (2) Robust-push restores reverse weights and introduces a stochastic future-pass stylisation sampler; and (3) SOTA-refine runs all losses at full weight.

IV. EXPERIMENTS

A. Protocol and Implementation

We build the protocol around the official COCO val2017 [6] split paired with the five exemplar paintings shipped with the AdaIN repository (*antimonocromatismo*, *impronte_d_artista*, *picasso_self_portrait*, *trial*, *woman_with_hat_matisse*). The complete 5 000-image val2017 archive is downloaded, shuffled with SEED=42 and then subsampled to 420 contents, so the selection is not biased towards any particular region of the dataset ordering. The result is split deterministically into 300 COCO contents $\times$ 5 styles = 1 500 training pairs and 120 *unseen* contents $\times$ 5 styles = 600 held-out evaluation pairs. At evaluation time the trained encoder/decoder pair runs in pure inference mode on each of the 600 unseen pairs: a fresh cover is rendered by $f_{\text{style}}(\mathbf{C}, \mathbf{S})$, the encoder embeds the semantic token, and the decoder is queried under the three modes of $\mathcal{A}$ (Section III-E); none of these pairs is seen during training, and any additional COCO image can be substituted at inference without retraining. Each pair is augmented with two precomputed serial covers (*direct-2*, *direct-3*) drawn from a deterministic rotation over the style list, so the serial-source loss can be evaluated without invoking f_{style} during training. Images are resized to 256×256 for the GPU benchmark and 192×192 for the CPU smoke run.

The four learnable modules are trained jointly with Adam [15], learning rate 6.5×10^{-5} , weight decay 10^{-5} and a cosine schedule, on a single *NVIDIA A100* rented through Google Colab in FP32 (mixed precision disabled), for 41 epochs. Training is not from scratch: it resumes from a serial-aware warm-start checkpoint produced by an earlier run of the same architecture, so exact reproduction requires that checkpoint in addition to the protocol described here. We release it with the code. One epoch over the 1 500 pairs takes ≈ 870 s, so the full schedule is ≈ 9.9 hours of wall-clock time. All loss hyper-parameters are listed in full for reproducibility.¹ The reported checkpoint was selected at epoch 39 with $V^* = 3.549$. The CPU smoke run uses the identical architecture and loss formulation at 192×192 with a fixed scalar batch (60 epochs,

¹ $\epsilon = 1.2 \times 10^{-4}$, $\lambda_{\text{cov}} = 25$, $\lambda_{\text{ceil}} = 30$, $\lambda_{\text{prior}} = 4.5$, $\lambda_{\text{sc}} = 0.06$, $\lambda_{\text{dwt}} = 0.2$, $\lambda_{\text{HH}} = 0.1$, $(\lambda_{\text{em-l}}, \lambda_{\text{em-t}}, \lambda_{\text{em-h}}) = (0.5, 0.3, 0.12)$, $(\lambda_{\text{r}}, \lambda_{\text{rR}}, \lambda_{\text{rI}}) = (25, 12, 12)$, $(\lambda_{\text{or}}, \lambda_{\text{od}}) = (2.6, 1.4)$, $\lambda_{\text{ser}} = 1.5$, $\beta = 0.7$, $(\lambda_{\text{tok}}, \lambda_{\text{tc}}, \lambda_{\text{th}}, \lambda_{\text{cs}}, \lambda_{\text{q}}) = (3.0, 0.25, 3.0, 1.6, 0.12)$, $\alpha = 0.025$, $p_{\text{f}} = 0.005$, $w_{\text{f}} = 0.04$.

one content/style pair) on a single CPU thread; it is the source of the CPU timings below.

B. Cover-Side Fidelity and Token Robustness

Table I reports cover-side fidelity and token-side robustness on the held-out partition as mean $\pm$ std over the 600 evaluation pairs at $\sigma_{\text{eval}} = 1.24$. The model attains 39.61 dB cover PSNR, 0.9914 SSIM and 0.0195 LPIPS, exceeding the 33–36 dB typically reported by style-transfer steganography approaches. The cover-PSNR standard deviation is only 0.35 dB, indicating that the capacity ceiling produces a stable empirical bound rather than a high-variance optimum. Robustness follows the same pattern: token MSE varies by at most 1×10^{-4} across the three attack modes, which translates into reverse SSIM under attack that is virtually indistinguishable from clean-channel decoding (Table II).

TABLE I
COVER-SIDE FIDELITY AND TOKEN-SIDE ROBUSTNESS ON THE COCO VAL2017 EVALUATION SPLIT (600 CONTENT-STYLE PAIRS, $\sigma_{\text{eval}} = 1.24$, BEST EPOCH 39). COVER MSE IS QUOTED AS A MEAN ONLY, SINCE ITS DISPERSION IS ALREADY EXPRESSED BY THE PSNR STANDARD DEVIATION.

Quantity	mean $\pm$ std
Cover MSE	1.10×10^{-4}
Cover SSIM	0.9914 ± 0.0030
Cover LPIPS	0.0195 ± 0.0081
Cover PSNR (dB)	39.6074 ± 0.3532
Token MSE (clean)	0.0133 ± 0.0043
Token MSE (RBN)	0.0133 ± 0.0045
Token MSE (JPEG)	0.0132 ± 0.0044

C. Reverse and Serial Style Transfer

Table II isolates the contribution of the semantic token to reverse recovery, comparing a content-free *naive reverse* baseline (which re-stylises the stego image back through f_{style} without access to any embedded payload) against four token-conditioned variants of our decoder. Reverse SSIM improves substantially, from 0.5234 to 0.8070 on the clean channel and to 0.8136 when the *oracle* variant is given the ground-truth token; PSNR improves correspondingly from 22.25 to 27.46 dB, a 5.21 dB gain attributable to the token alone. The token-conditioned decoder is largely insensitive to the attack it meets at inference time — SSIM varies only from 0.8070 (clean) to 0.7884 (Gaussian) and 0.7898 (JPEG) — and LPIPS moves in the same direction, from 0.189 to 0.122, so the improvement is perceptual as well as numerical.

Table III reports recovery after one extra re-stylisation (*serial-2*) and two extra re-stylisations (*serial-3*), and two consistent patterns emerge. First, SSIM is the only metric on which the naive baseline slightly exceeds the proposed method (0.4966 vs. 0.4943 at *serial-2*); the margin is small and reversed on every other metric — L2 falls from 0.0090 to 0.0078, LPIPS from 0.1945 to 0.1577 and PSNR rises from 20.64 to 21.41 dB — so token-conditioned recovery wins on three of four metrics at both serial steps. Second, the token-conditioned variants are markedly more *stable*: their SSIM standard deviation is consistently lower than the naive

TABLE II

REVERSE STYLE TRANSFER COMPARISON (600 EVALUATION PAIRS, $\sigma_{\text{EVAL}} = 1.24$). *naive_reverse* IS A CONTENT-FREE RE-STYLISATION BASELINE; *token_** VARIANTS USE OUR REVERSE DECODER. *oracle* READS THE GROUND-TRUTH TOKEN, THE OTHERS OPERATE ON THE DECODED TOKEN AFTER IDENTITY / RBN / JPEG DISTORTION.

Method	L2 ↓	SSIM ↑	LPIPS ↓	PSNR ↑
naive_reverse	0.0065 ± 0.0029	0.5234 ± 0.0852	0.1889 ± 0.0905	22.2
token_reverse_clean	0.0022 ± 0.0015	0.8070 ± 0.0858	0.1216 ± 0.0799	27.4
token_reverse_rbn	0.0027 ± 0.0019	0.7884 ± 0.0931	0.1514 ± 0.0893	26.5
token_reverse_jpeg	0.0025 ± 0.0018	0.7898 ± 0.0968	0.1467 ± 0.0909	26.9
token_reverse_oracle	0.0021 ± 0.0015	0.8136 ± 0.0852	0.1135 ± 0.0768	27.6

baseline’s. For a steganographic system this is a favourable trade: a marginal reduction in SSIM buys substantially more predictable output. The Gaussian-channel (*token_rbn*) and blended serial-source (*blend_**) variants are omitted here for space; they follow the same ordering, winning on L2, LPIPS and PSNR while trailing the naive baseline on SSIM.

TABLE III

SERIAL STYLE TRANSFER AFTER ONE (*serial-2*) AND TWO (*serial-3*) ADDITIONAL RE-STYLISATIONS, 600 PAIRS. *TOKEN_** ROWS USE OUR REVERSE DECODER; *NAIVE* IS THE CONTENT-FREE BASELINE.

Method	L2 ↓	SSIM ↑	LPIPS ↓	PSNR ↑
naive	0.0090 ± 0.0027	0.4966 ± 0.0936	0.1945 ± 0.0725	20.64 ± 1.65
<i>serial-2</i> token_clean	0.0078 ± 0.0030	0.4943 ± 0.0844	0.1577 ± 0.0585	21.41 ± 1.65
token_jpeg	0.0081 ± 0.0031	0.4796 ± 0.0831	0.1662 ± 0.0591	21.23 ± 1.89
naive	0.0113 ± 0.0033	0.4548 ± 0.0767	0.2378 ± 0.0684	19.64 ± 1.27
<i>serial-3</i> token_clean	0.0091 ± 0.0035	0.4500 ± 0.0881	0.1885 ± 0.0627	20.71 ± 1.85
token_jpeg	0.0094 ± 0.0036	0.4376 ± 0.0872	0.1984 ± 0.0635	20.58 ± 1.70

D. Ablation of the Loss Terms

To isolate each design choice we ran a separate controlled ablation at the reduced 192×192 CPU resolution. Removing the capacity ceiling reduces cover PSNR by 5.76 dB ($41.37 \rightarrow 35.61$ dB): the network still encodes the token successfully but no longer guarantees a high-quality cover. The frequency loss in isolation is essentially neutral with respect to cover PSNR because it does not bound the residual magnitude, only its spectral allocation. With both losses enabled, cover PSNR returns to the ceiling-only regime while the residual concentrates in the high-frequency band.

E. Qualitative Inspection and Cross-Dataset Generalisation

Figure 2 shows qualitative output at 256×256 on four representative images drawn from COCO, DIV2K, CelebA and BOSSBase. Cover and stego are visually indistinguishable to the naked eye. The encoder is also not tuned to a single artistic texture: the 0.35 dB standard deviation of cover PSNR in Table I already bounds the across-style variation.

Cross-dataset evaluation on COCO, DIV2K, CelebA, and BOSSBase yields consistent cover PSNRs between 39.4–40.6 dB, indicating that the capacity ceiling and frequency loss carry over beyond the training content–style pairs. Token MSE is computed by a standalone decoder; its relative stability across attacks remains the operative robustness metric.

Table IV presents the robustness evaluation under various attacks. Token MSE is virtually unchanged across all standard



Fig. 2. Qualitative behaviour across COCO, DIV2K, CelebA, and BOSSBase. Columns: original content **C**, artistic style **S**, stylised cover **X_{cover}**, final stego image **X_{stego}**.

image-processing attacks — JPEG compression, Gaussian noise, resizing, and cropping. The resilience extends to serial re-stylisation: passing the stego image through NST with completely different styles one or two further times raises Token MSE by merely ~ 0.01 .

TABLE IV
ROBUSTNESS (TOKEN MSE Δ VS ID).

Attack	Avg MSE	Δ
Identity	0.3602	—
JPEG Q50	0.3598	−0.0004
Gaussian $\sigma=0.05$	0.3600	−0.0002
Resize 50%	0.3600	−0.0002
Crop 80%	0.3617	+0.0015
Serial-2 [†]	0.3716	+0.0114
Serial-3 [†]	0.3706	+0.0104

F. Comparison Against Prior Art

Table V compares our method against the style-transfer steganography family; only methods reporting concrete numbers are listed, with purely qualitative ones discussed in Section II. Because different works measure PSNR against different reference images (pre-style cover, post-style cover, or recovered secret), we state the *metric target* explicitly rather than let the numbers imply a false equivalence: Quaternion-Style [11] (42.30 dB) evaluates PSNR against the *pre-style* natural cover, a strictly easier target, and IHST [4] (35.90 dB) measures *secret recovery* rather than cover preservation. Our model is nonetheless the only one to simultaneously report (i) reverse SSIM above 0.80, (ii) robustness to JPEG and Gaussian attacks, and (iii) serial re-stylisation survival across three cumulative stylisation passes. For broader context, general deep steganography reports container PSNR up to 40.49 dB (RIIS [2]), while coverless methods report extraction accuracy

above 97% (IDEAS [10]); these payloads differ fundamentally from ours, so the numbers are context rather than comparison.

TABLE V

COMPARISON WITHIN THE STYLE-TRANSFER STEGANOGRAPHY FAMILY. ONLY METHODS REPORTING CONCRETE PSNR/SSIM VALUES ARE INCLUDED. NUMBERS ARE VERBATIM FROM EACH ORIGINAL PAPER; CROSS-ROW COMPARISON REQUIRES CAUTION DUE TO PROTOCOL DIFFERENCES.

Method	Year	PSNR $\uparrow$	SSIM $\uparrow$	Rev.
Self-Contained [3]	2020	—	—	✓
Quaternion [11]	2021	42.30	0.987	—
IHST [4]	2024	35.90	0.960	✓
Ours	2026	39.61	0.991	0.81

Metric target: Self-Contained reports L2/SSIM/LPIPS with no PSNR;

Quaternion measures stego $\leftrightarrow$ *pre-style* cover; IHST measures recovered $\leftrightarrow$ secret (not cover fidelity); Ours measures stego $\leftrightarrow$ *styled* cover.

V. CONCLUSION

This paper identified two design choices — an unbounded capacity penalty and a spatial payload representation — that constrain existing style-transfer steganography, and proposed FGSS to overcome them. A $64 \times 32 \times 32$ semantic token replaces the spatial secret, and a frequency loss coupled with a soft capacity ceiling directs the residual into the high-frequency band. An oracle-guided reverse extraction stack further stabilises decoding under standard attacks and across repeated re-stylisation. Experimental results on COCO validate our approach, achieving 39.61 dB cover PSNR and reverse SSIM of 0.8070, outperforming prior baselines in both robustness and serial survivability. Future work will investigate adversarial steganalysis explicitly during training to close the remaining detectability gap.

REFERENCES

- [1] Lu, S.-P. et al.: Large-capacity image steganography based on invertible neural networks. In: CVPR (2021)
- [2] Xu, Y. et al.: Robust invertible image steganography. In: CVPR (2022)
- [3] Chen, H.-Y. et al.: Self-contained stylization via steganography. In: WACV (2020)
- [4] Zhang, F. et al.: Conditional image hiding network based on style transfer. Information Sciences **662**, 120225 (2024)
- [5] Lin, Y.-H. et al.: A steganographic message transmission method based on style transfer and DDPM. Electronics **14**(16), 3258 (2025)
- [6] Lin, T.-Y. et al.: Microsoft COCO: common objects in context. In: ECCV (2014)
- [7] Wang, Z. et al.: Image quality assessment. IEEE TIP **13**(4), 600–612 (2004)
- [8] Zhang, R. et al.: The unreasonable effectiveness of deep features as a perceptual metric. In: CVPR (2018)
- [9] Boroumand, M. et al.: Deep residual network for steganalysis. IEEE TIFS **14**(5), 1181–1193 (2019)
- [10] Liu, X. et al.: Image disentanglement autoencoder for steganography without embedding. In: CVPR (2022)
- [11] Li, Q. et al.: Image steganography based on style transfer and quaternion exponent moments. Applied Soft Computing **110** (2021)
- [12] Gatys, L.A. et al.: Image style transfer using CNNs. In: CVPR (2016)
- [13] Simonyan, K., Zisserman, A.: Very deep convolutional networks. In: ICLR (2015)
- [14] Hendrycks, D., Gimpel, K.: Gaussian error linear units (GELUs). arXiv:1606.08415 (2016)
- [15] Kingma, D.P., Ba, J.: Adam: a method for stochastic optimization. In: ICLR (2015)